

Sum Yan Wan | Charmaine

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SUMMARY

Computer Engineering student at the University of Ottawa with hands-on experience in software development, AI/ML, web applications, and database design.

Experienced in building real-time systems using Python, Java, JavaScript, Firebase, and machine learning models, with strong interests in intelligent applications, automation, and full-stack development.

EDUCATION

BASc Computer Engineering, Cum Laude

University of Ottawa, Ontario, Canada

2022–2026

Academic recognition: Dean's Honour List; Merit Scholarship recipient

PROJECTS

ECOLlector — Smart Recycling System | Software Developer

2025–2026

- Built software for real-time waste detection, classification, and routing.
- Developed the MobileNetV4 model, Firebase database, and main web app.

Local CPU-Only Voice Assistant

2025–Present

- Built a local voice assistant using Tacotron 2, Vosk, and Qwen2.5.
- Added intent detection, cached embeddings, auto-reload, and GUI settings.

Tutron App

2023

- Developed an Android tutoring app using Java, Android Studio, and Firebase.
- Applied OOP, unit testing, and CircleCI CI/CD in a team project.

EXPERIENCE

Lighting & Shading Assistant (Paid Trainee)

Mr Meal, Hong Kong

June 2021 - July 2021

- Worked as part of a collaborative art team to produce 3D-rendered assets for a 2D game.
- Set up lighting, shading, and UVs to produce high-quality renders for 2D integration.
- Developed strong teamwork and communication skills in a professional studio environment.

TECHNICAL SKILLS

Programming Languages: C, C++, C#, Java, Python, GoLang, Prolog, Scheme

AI/ML: MobileNetV4, Tacotron 2, Vosk, Qwen2.5, sentence embeddings, cosine similarity

Web & Mobile Development: JavaScript, TypeScript, React, Android Studio, CSS, HTML

Backend&Database: Firebase, CI/CD, MySQL, SQLite, PostgreSQL

Tools: Maya, Adobe Photoshop, Godot, Docker

3D Art: Lighting, shading, UV Mapping, basic Modeling

Spoken Languages: English, Cantonese, Mandarin

HONOURS & AWARDS

Merit Scholarship, University of Ottawa — 2023, 2025

Dean's Honour List, University of Ottawa — Summer 2023, Fall 2025, Winter 2026